- Suitable for data authentication applications, compliant with:
  - Federal Information Processing Standards Publication FIPS PUB 180-4, Secure Hash Standard (SHA-1 and SHA-2 family)
  - Federal Information Processing Standards Publication FIPS PUB 186-4, Digital Signature Standard (DSS)
  - Internet Engineering Task Force (IETF) Request For Comments RFC 2104, HMAC: Keyed-Hashing for Message Authentication and Federal Information Processing Standards Publication FIPS PUB 198-1, The Keyed-Hash Message Authentication Code (HMAC)
- Support for HMAC mode with all supported algorithms
- Corresponding 32-bit words of the digest from consecutive message blocks are added to each other to form the digest of the whole message.
  - Automatic 32-bit word swapping to comply with the internal little-endian representation of the input bit-string
  - Supported word swapping format: bits, bytes, half-words, and 32-bit words
- Single 32-bit, write-only, input register associated with an internal input FIFO, corresponding to a 64-byte block size (16 × 32 bits)
- Automatic padding to complete the input bit string to fit the digest minimum block size
- AHB slave peripheral, accessible by 32-bit words only (else an AHB error is generated)
- 8 × 32-bit words (H0 to H15) for output message digest
- Automatic data flow control supporting direct memory access (DMA) using one channel
- Support for both single and fixed DMA burst transfers of four words
- Interruptible message digest computation, on a per-block basis
  - Reloadable digest registers
  - Hashing computation suspend/resume mechanism, including DMA

3.21 Timers and watchdogs

The devices include two advanced control timers, up to three general-purpose timers, two basic timers, one low-power timer, two watchdog timers, and one SysTick timer.

The table below compares the features of the advanced control, general-purpose and basic timers.

Table 5. Timer feature comparison

Timer type	Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture / compare channels	Complementary outputs
Advanced control	TIM1, TIM8	16 bits	Up, down, Up/ down	Any integer between 1 and 65536	Yes	4	4
General-purpose	TIM2	32 bits	Up, down, Up/ down	Any integer between 1 and 65536	Yes	4	No
	TIM12,	16 bits	Up			2	
	TIM15	16 bits	Up, down, Up/ down			2	
Basic	TIM6, TIM7	16 bits	Up	Any integer between 1 and 65536	Yes	0	No

3.21.1 Advanced-control timers (TIM1/TIM8)

The advanced-control timers (TIM1/TIM8) consist of a 16-bit autoreload counter driven by a programmable prescaler.

They may be used for various purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

TIM1/TIM8 timer features include: